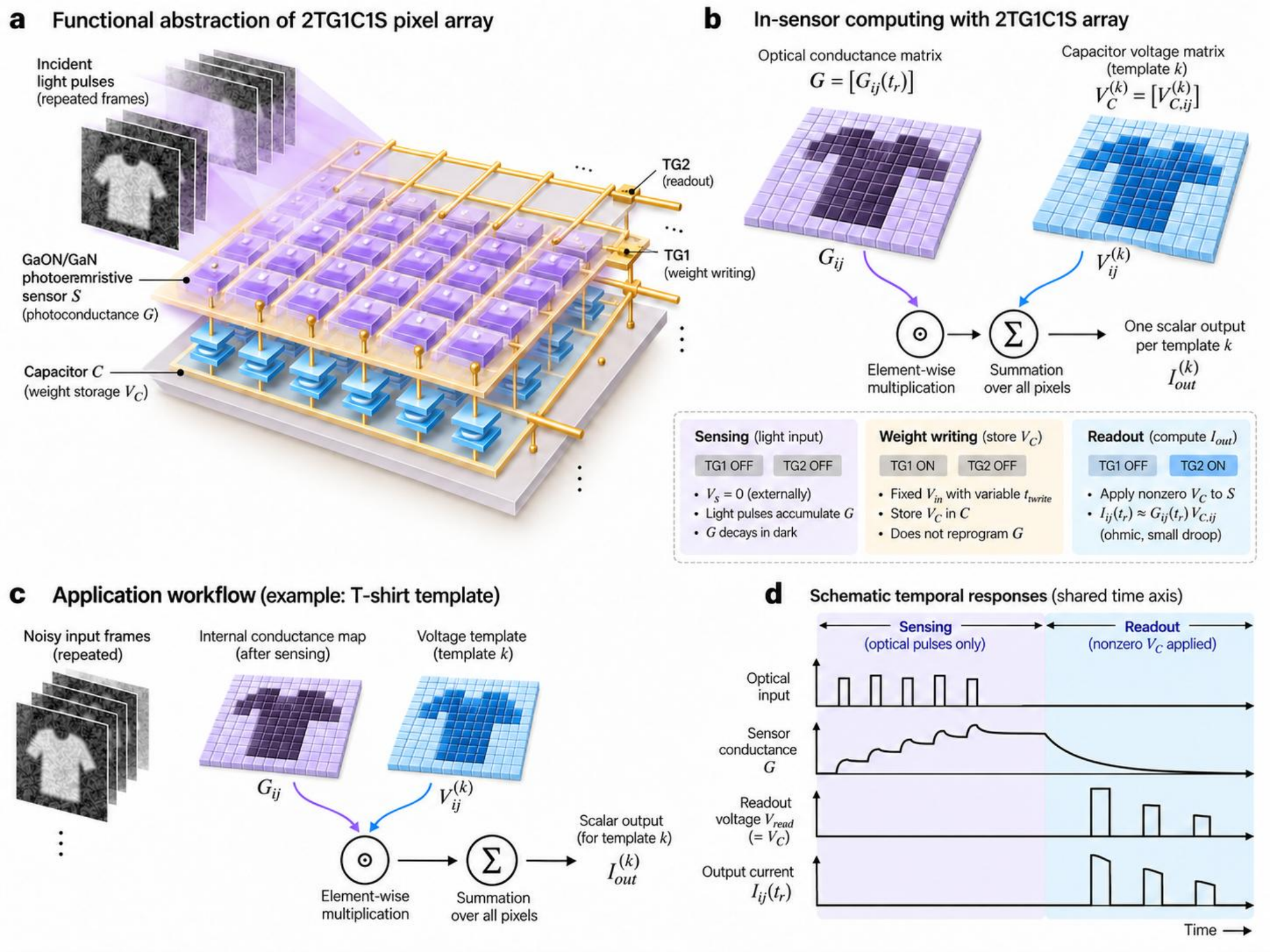

a Functional abstraction of 2TG1C1S pixel array
Incident light pulses (repeated frames)
GaON/GaN photoermristive sensor S (photoconductance G)
Capacitor C (weight storage V_C)
TG2 (readout)
TG1 (weight writing)
b In-sensor computing with 2TG1C1S array
Optical conductance matrix
$G = [G_{ij}(t_r)]$
Capacitor voltage matrix (template k)
$V_C^{(k)} = [V_{C,ij}^{(k)}]$
G_{ij}
$V_{ij}^{(k)}$
Element-wise multiplication
Summation over all pixels
One scalar output per template k
$I_{out}^{(k)}$
Sensing (light input)
TG1 OFF
TG2 OFF
$V_s = 0$ (externally)
Light pulses accumulate G
G decays in dark
Weight writing (store V_C)
TG1 ON
TG2 OFF
Fixed V_{in} with variable t_{twrite}
Store V_C in C
Does not reprogram G
Readout (compute I_{out})
TG1 OFF
TG2 ON
Apply nonzero V_C to S
$I_{ij}(t_r) \approx G_{ij}(t_r)\, V_{C,ij}$
(ohmic, small droop)
c Application workflow (example: T-shirt template)
Noisy input frames (repeated)
Internal conductance map (after sensing)
Voltage template (template k)
G_{ij}
$V_{ij}^{(k)}$
Element-wise multiplication
Summation over all pixels
Scalar output (for template k)
$I_{out}^{(k)}$
d Schematic temporal responses (shared time axis)
Sensing (optical pulses only)
Readout (nonzero V_C applied)
Optical input
Sensor conductance G
Readout voltage V_{read} (= V_C)
Output current $I_{ij}(t_r)$
Time